Ranuga Amarasinghe

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Sri Lanka

EDUCATION

Dharmapala College, Pannipitiya, Sri Lanka | *(English Medium)*

- **GCE Ordinary Level** | 8 A grades and 1 C grade

  A grades: Science, Mathematics, English, Buddhism, Sinhala, History, ICT, Business & Accounting Studies. | C: English Literature.
- **GCE Advanced Level** - Combined Mathematics, Physics, Chemistry | Sat August 2026; results pending
- **GCE Advanced Level** - General Information Technology (GIT): A Grade.

SELECTED PROJECTS

- **Multitasking Hexacopter** [↗] **-** Built a hexacopter capable of complex maneuvers and autonomous missions, including school-flag demonstrations at events.
- **AUV for SAUVC 2024 (HEX-DIVE)** [↗] **-** Developed a Raspberry Pi- and Arduino-based autonomous underwater vehicle with navigation, sensor integration, and underwater task execution.
- **Arduino Flight Controller-Based Quadcopter** [↗] **-** Built a quadcopter using an Arduino-based flight control system.
- **SKY-SOW - Multitasking Agricultural Drone** [↗] **-** Developed a Raspberry Pi- and Arduino-based agricultural drone that won third place in the National Robotics and Innovation Competition organized by the Ministry of Education, Sri Lanka.
- **HEX-BOTZ - SLRC 2022/23** [↗] **-** Designed and built an autonomous robot as part of the HEX-BOTZ team for the Sri Lankan Robotics Challenge, organized by the University of Moratuwa.
- **ROV V1 - Underwater Robot** [↗] **-** Built an Arduino-controlled, three-thruster tethered underwater robot that won second prize in Instructables "Make It Move" contest.
- **Micromouse Maze-Solving Robot** [↗] **-** Built an autonomous robot that navigates and solves mazes; achieved second place in the WARRIOR BOTS competition.
- **Functional Model Plane** [↗] **-** Built a functional trainer aircraft, exploring aerodynamics and flight systems.

TECHNICAL SKILLS

- **Programming & App Development:** Python, C++, Dart, Flutter, HTML, PHP
- **Hardware & Prototyping:** Arduino, Raspberry Pi, PCB design
- **CAD & 3D Design:** Autodesk Fusion 360, SOLIDWORKS, 3D Printing

HONORS & AWARDS

- **Merit Award - 19th Sri Lankan Physics Olympiad Competition (SLPhO) 2025** - Scored 77% in the national physics competition.
- **2nd Place - Sri Lankan Robotics Challenge (SLRC) 2024** - University of Moratuwa, Mar 2024 | Robotics design and implementation.
- **3rd Place - All Island Robotics and Innovation Competition** - Ministry of Education, Sri Lanka, Dec 2023 | Developed agricultural drone.
- **2nd Place - Maze Solver Robot (Micromouse "WARRIOR BOTS" Competition)** - CINEC Campus, Jul 2023 | Built maze solving robot.
- **Grand Prize - HEX-DIVE ("Water" Contest)** - Instructables.com, Aug 2024 | Won with an underwater robotics project.
- **Second Prize - ROV V1 ("Make It Move" Contest)** - Instructables.com, Sep 2021 | Built a tethered underwater robot.
- **Runner-Up - Make It Real Student Design Challenge #3** - Instructables.com, Jun 2021 | Created a 3D CAD model of a fitness center.

LEADERSHIP EXPERIENCE

- **President - Young Inventors' Club** - Dharmapala College, Pannipitiya | Jan 2023 - Dec 2024: Led STEM activities, mentored students, and coordinated innovation projects.
- **Chief Organizer - Senior Science Society** - Dharmapala College, Pannipitiya | Jan 2025 - Jan 2026: Organized science events, coordinated teams, and promoted STEM engagement.
- **Team Leader** - HEX-BOTZ, Sri Lankan Robotics Challenge (2022/23) — University of Moratuwa.
- **Team Leader** - HEX-BOTZ, MechatronX (2023) — University of Moratuwa.
- **Team Leader** - HEX-BOTZ, WARRIOR BOTS (2023) — 2nd Place, CINEC University.
- **Team Leader** - HEX-DIVE, SAUVC (2024) — International underwater robotics competition.
- **Team Leader** - HEX-BOTZ, Sri Lankan Robotics Challenge (2024) — 2nd Place, University of Moratuwa.

LANGUAGES

English - Professional proficiency | **Sinhala** - Native